TENZIN LHAKPA

AI / ML Engineer — Agentic Systems · LLM Applications · Production ML
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EDUCATION

Master of Science, Data Science and Applications Dec 2024
University at Buffalo, The State University of New York, USA

Bachelor of Engineering, Computer Science and Engineering Oct 2020
Rajiv Gandhi Proudyogiki Vishwavidyalaya University, India

WORK EXPERIENCE

AI / Machine Learning Engineer II | PrecisionAQ May 2025 – Present

SherpaAI — Multi-Agent EHR Automation

- Designed and built the multi-agent architecture for SherpaAI as a LangGraph-orchestrated network of five AI agents — with LangChain underneath — that autonomously extracts and structures drug protocols for EHR readiness.
- Powered agent reasoning with Azure OpenAI and instrumented the system with LangSmith for tracing, debugging, and evaluation of multi-agent runs.
- Reduced manual effort for EHR technical experts by 50–60%.

CoverageRX

- Architected an end-to-end data pipeline leveraging Azure OpenAI and Microsoft Agentic AI to extract and standardize drug-coverage data from annually published formulary documents across multiple providers at scale.
- Engineered and deployed a production-grade agentic RAG chatbot for the ARD team, enabling natural-language retrieval of complex insights from Snowflake-backed datasets and eliminating dependency on manual data queries.
- Built an autonomous HCP research agent that aggregates, validates, and stores web-sourced provider data, boosting research-team productivity by 30% through significant reduction in manual effort.

MatrixAI

- Designed and integrated a tweet-relevance scoring mechanism into the MatrixAI pipeline using AI-based filtering, reducing GPT-5 API calls by 27% and meaningfully lowering token usage and compute cost.
- Developed and deployed an Azure Container App for scraping drug-related information, supported by Playwright.
- Developed and deployed Azure Functions and Azure Queues to process tweets, generating AI-based sentiment and training a BERTopic model to surface tweet topics.

PatientLens

- Optimized the ML-based imputation algorithm for PatientLens, cutting execution time by 45% and saving ~$4,200 annually across monthly production runs.

Machine Learning Engineer | Machine Motoring Systems LLC Sept 2024 – Apr 2025

- Engineered and deployed an end-to-end ML pipeline for real-time fault detection in rotary machines, reducing maintenance costs by 80%.
- Pre-trained a ResNet18-based encoder on FFT and time-series vibration data, leveraging transfer learning and a self-supervised framework with a cross-correlation matrix as the loss function.
- Developed the data pre-processing and data-engineering pipeline, enabling more efficient analysis.
- Performed fault simulations on an STI Rotorkit, capturing the unique vibration patterns each fault generates.
- Achieved 98% accuracy, paving the way for commercial deployment.

Machine Learning Research Assistant | SUNY Buffalo July 2024 – May 2025

- Created EndoAssistant, a multimodal vision-language system that provides guidance on surgical procedures.
- Implemented QLoRA-based parameter-efficient fine-tuning on LLaVA-7B with custom medical datasets, optimizing training through DeepSpeed ZeRO-3 for resource-efficient training on limited GPU infrastructure.
- Developed and curated a comprehensive vision-language dataset for endoscopic surgery, spanning 1,000+ annotated surgical videos and 5,000+ specialized medical resources.
- Built an ML pipeline to detect cancerous cells from whole-slide-image (WSI) medical images.
- Applied a convolutional GRU fusion model for image pre-processing and feature fine-tuning.
- Built and optimized the data-collection pipeline, reducing processing time by 30%.
- Implemented OpenAI's CLIP model to classify 100K+ medical images using advanced prompt-engineering techniques, achieving 80% accuracy.

Database Manager | CTA, India Jan 2021 – Jan 2023

- Managed CTA's largest database, ensuring accuracy and efficiency.

- Architected and implemented a digital identity system for Tibetans in exile, managing records for 100,000+ individuals.
- Led a 45-member team executing a $43K USD project, resulting in an 80% increase in policy-implementation efficiency.
- Directed developers and officers in building a new database system for the Tibetan Demographic Survey Project.

KEY PROJECTS

MunSel: Agentic Tibetan Language-Learning Platform | Monlam AI Hackathon 2026

- Built MunSel, a production language-learning platform unifying an agentic tutor, speech-based practice drills, and an autonomous bilingual newsroom in a single FastAPI service — deployed on Railway with a React + TanStack Router frontend.
- Engineered a LangGraph tool-calling research agent that grounds an LLM tutor: a GPT-4.1-mini orchestrator retrieves verified vocabulary from three prioritized sources (curated phrasebook, dictionary API, OCR-built lexicon) before a separate generation model composes the reply — a retrieval/generation split that eliminated the hallucinated-vocabulary failures of a single-model ReAct loop.
- Designed an agentic news pipeline that concurrently crawls 8 RSS feeds (HTTP-304-aware, with deduplication and feed-turnover detection) and applies two-stage relevance screening — deterministic rule filters plus batched LLM adjudication at 15 articles per call — then clusters events across English and Tibetan, ranks by cross-outlet consensus, and auto-generates a sectioned bilingual digest.
- Implemented in-browser speech features: WAV PCM capture via the Web Audio API for speech-to-text pronunciation grading, punctuation-tolerant transcript normalization, and real-time stroke-order grading across 34 Tibetan glyphs (136 strokes).

Tech Stack: Python, FastAPI, LangChain, LangGraph, GPT-4.1-mini, React, TanStack Router, SQLite, Docker, Railway, RAG, TTS/STT, OCR.

JoBot: AI Agent for Job Applications | Personal Project

- Engineered a dual-agent system — a Forger (web scraper) and an Applier (form-filling agent) — built on Playwright browser automation and the Kimi K2 multimodal LLM via an OpenAI-compatible API.
- The Forger scrapes LinkedIn job listings using a CSS-selector fast path with a Claude Sonnet vision fallback, exporting structured results to Excel via openpyxl.
- The Applier runs a multi-turn agentic loop with 8 browser-interaction tools (screenshot, HTML extraction, form filling, file upload, DOM navigation), persistent agent memory via Pydantic-validated structured tool calls, and an ATS knowledge base for Greenhouse, Lever, Workday, Keka, and Ashby; fallback click strategies (direct, JS, ActionChains) handle complex DOM structures for seamless navigation across diverse sites.
- Designed a token-efficient context-management strategy, replacing full conversation history with a compact rolling memory notebook and reducing per-turn token cost by ~80%.
- Integrated BeautifulSoup HTML parsing to extract structured form metadata (inputs, selects, textareas, buttons with precomputed CSS selectors), eliminating redundant page-HTML calls.
- Implemented bot-detection mitigations including randomized hover/click/type timing, persistent browser profiles, and navigator.webdriver suppression via Playwright init scripts.

AIHeadShotX: Generative AI Portrait SaaS | Personal Project

- Designed a multimodal inference pipeline using the Google Gemini API that conditions portrait generation on three facial reference images (front/left/right) plus an optional style reference, achieving consistent identity preservation across 7 style presets.
- Engineered prompt-engineering strategies for style-conditioned image synthesis (corporate, editorial, style-match), iterating on structured prompts to control lighting, composition, and subject fidelity.
- Built a tiered output-quality system — free tier receives downscaled, watermarked outputs; paid tiers receive full-resolution generations — implemented via PIL post-processing.
- Productionized the model behind an async FastAPI inference server with request queuing, error handling, and credit deduction only on successful generation (preventing loss on model failure).
- Deployed end-to-end on Railway with a PostgreSQL backend, Cloudflare R2 artifact storage, and automated expiry cleanup — serving real users with a React + Vite frontend.

AI-Powered Medical Document Processing Pipeline | Freelance Project

- Built an AI-powered medical document processing pipeline that automated extraction of structured clinical data from millions of hospital PDFs stored in AWS S3, reducing reliance on manual data-entry workflows.
- Developed a hybrid OCR architecture using PyMuPDF and Tesseract OCR with adaptive output selection to improve extraction accuracy across scanned and digital medical records.
- Integrated Gemini 2.5 Flash for LLM-based medical information extraction, transforming unstructured healthcare documents into structured PostgreSQL-compatible schemas.
- Increased structured healthcare database coverage by 40% by automating large-scale ingestion of previously unprocessed records, requiring only ad-hoc human verification.

Tech Stack: Python, AWS S3, PyMuPDF, Tesseract OCR, Gemini 2.5 Flash, PostgreSQL, ETL, Document AI.

TECHNICAL SKILLS

GenAI & Agentic: LangGraph, LangChain, LangSmith, Multi-Agent Systems, Agent Orchestration, RAG, Agentic Workflows, Azure OpenAI, OpenAI API, Claude API, Google Gemini, Hugging Face Transformers, QLoRA / PEFT, DeepSpeed, CLIP, LLaVA, BERTopic, Stable Diffusion

ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Languages: Python, R, SQL, C, C++, Java

Data & Cloud: Azure (OpenAI, Functions, Queues, Container Apps), AWS (S3), Snowflake, PostgreSQL, SQLite, Vector Databases, ETL Pipeline Design

Backend & Web: FastAPI, React, Vite, TanStack Router, Playwright, REST APIs, HTML, CSS

MLOps & Tools: Docker, Git, Railway, Cloudflare R2

KEY ACHIEVEMENTS

- Fulbright Scholarship recipient — awarded by the U.S. Government for academic excellence and leadership potential.
- Engineering Reserved Seat Scholarship recipient from the Tibetan Government.